

3. How much should the manager invest in the stock?

Part 1. How much alpha can a manager capture if the order size is 10% ADV?

The solution to how much alpha can a manager capture if the order size is 10% ADV is found by maximizing our expected profit (Equation 11.23). In this example, the alpha forecast is 3% over three days. For simplicity, we illustrate this concept using a linear appreciation model. In practice, managers can incorporate any trend preference they have, such as a compounded model, non-linear, exponential, or even a step function where the return only occurs overnight and is constant during the day.

The alpha capture optimization is illustrated in Figure 11.9 for a small cap stock with annualized volatility = 43%. If we execute an order of 10% ADV ultra-aggressively the market impact cost will be 132 bp (the instantaneous impact cost) but we will not incur any alpha cost. Since the stock will return 3% or 300 bp over the period this urgent execution strategy will earn us a net return of 168 bp. If we trade passively over the entire three day period our impact cost will be 23 bp but our alpha cost will be 150 bp. The total cost will be 173 bp and our net profit will be 127 bp which is less than trading the entire position at once. A naïve analyst may elect to trade the entire order at once to earn a higher expected return. But this would not be an appropriate option. Take a look at the graph in Figure 11.9. Notice how the market impact cost is decreasing over the period and alpha cost is increasing over the period as expected. But most importantly, take a look at the total cost curve. This cost starts high when market impact is dominating the total cost, decreases, and then begins to increase again when the alpha cost starts to dominate the total cost. The total cost function will always be a convex function unless the manager is buying stocks that are decreasing in value or selling stocks that are increasing in value.

The minimum cost occurs at a trading time of 0.45 days (just slightly less than half of a day in volume time). The corresponding market impact cost is 54 bp and corresponding alpha cost is 23 bp for a total cost of 77 bp. The net profit the manager can earn is 223 bp ($300 \text{ bp} - 77 \text{ bp} = 223 \text{ bp}$). Notice that this is much larger than the profit of 168 bp for the ultra-aggressive instantaneous strategy and the 127 bp for the passive strategy. The maximum alpha capture for this order is 223 bp.

Part 2. How much can be invested in this stock before we begin to incur a loss is found by maximizing the number of shares that be transacted at